TITLE PAGE

Digital Literacy: Student Digital Portfolio

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INTRODUCTION

This Digital Literacy Project was assigned as part of the CSE0001 course to help us understand how to use digital tools effectively and responsibly as students. In this project, I worked as a Student Digital Ambassador whose role is to support classmates in building a professional online presence and improving safe communication on digital platforms.

Through this project, I completed five different tasks such as creating a digital literacy infographic, setting up profiles on professional platforms like GitHub and LinkedIn, exploring coding and collaboration tools, learning proper email etiquette, and preparing a cybercrime awareness case study. These activities helped me improve my practical digital skills and understand the importance of using technology safely and professionally.

Overall, this project was a useful learning experience for my academic journey and future career. 💻📬✨

Task 1 Create a Digital Literacy Awareness Infographic

Tool Used and Design Concept

I utilized **Canva**, an intuitive online graphic design platform, to create the "Digital Literacy Awareness Infographic." This tool was selected for its extensive library of vector icons, professional templates, and flexible layout controls, which simplified the process of condensing complex information into a visually digestable format.

The infographic is structured around the core "Pillars of Digital Literacy." It uses a clear, four-quadrant layout that visually guides the viewer through the key components of navigating the online world safely. The top half defines **Information Evaluation** and **Critical Thinking**, illustrating the process of fact-checking and recognizing bias using icons of magnifying glasses and question marks. The bottom half transitions into the behavioral aspects: **Secure Online Behavior** (featuring locks and shields to emphasize password security) and **Digital Etiquette** (representing respectful communication through a megaphone icon). The overall design uses a clean, modern aesthetic with a blue and yellow color palette to symbolize trust, intelligence, and accessibility, ensuring the content is engaging rather than intimidating.

A Challenge in Creation

One of the main difficulties I encountered was managing the **information hierarchy**. It was initially challenging to take comprehensive concepts, like the legal ramifications of cybercrime discussed in previous sections, and distill them into just three or four brief bullet points per quadrant without sacrificing the core message. I had to focus on simplifying technical language—for example, replacing a detailed explanation of multi-factor authentication with the immediate behavioral instruction "Use Strong Passwords"—to maintain the readability and immediate impact necessary for an effective public awareness infographic.

Task 2 Build Your Student Digital Portfolio**Selection and Strategy: My Digital Portfolio Ecosystem**

To build a robust professional presence over the next four years, I have selected **LinkedIn**, **GitHub**, and **Google Sites** as the primary pillars of my digital portfolio. Each platform serves a distinct functional purpose that, when combined, creates a comprehensive narrative of my growth and technical expertise.

- **LinkedIn** will function as my professional networking hub and "living resume." I intend to use it to document career milestones, share insights from my coursework, and connect with industry leaders in criminal justice and forensics.
- **GitHub** is my designated repository for technical proof-of-work. It will house my code for digital forensic tools and software development projects, providing a transparent look at my version control habits and problem-solving logic.
- **Google Sites** will act as the master curator. It provides a polished, visual interface to host my case studies, academic presentations, and long-form reflections that don't fit into a standard code repository.

Over the next four years, I plan to update these platforms at the end of every semester. This consistent cadence will allow me to track the evolution of my skills—moving from fundamental concepts to complex, independent investigations—ensuring that by graduation, I possess a data-driven record of my academic and professional journey.

Task 3 Explore Coding & Collaboration Platforms

Part B – Google Workspace Collaboration (mandatory):**Collaboration and Implementation: Navigating the Digital Workflow**

For this project, **Google Workspace** served as the mandatory foundation for collaboration, supplemented by **GitHub** for technical version control. Google Docs and Drive were the primary environments for drafting the project report and organizing case studies, while GitHub provided a centralized location for hosting the scripts and configuration files related to my forensic tool setups.

The integration of these platforms was not without its hurdles. The primary challenge I faced was **maintaining version parity** between my local development environment and the cloud-based collaborative tools. While Google Workspace excels at real-time text editing, it lacks native support for rendering code or tracking terminal-based configurations. I often had to manually bridge the gap by exporting logs and screenshots from my terminal into the shared documents, which occasionally led to formatting inconsistencies.

Additionally, managing **permissions and access control** required a steep learning curve. Ensuring that all project components—from private repository snippets to shared presentation slides—were accessible to the right stakeholders without compromising security was a delicate balance. Overcoming these obstacles ultimately improved my technical literacy, teaching me how to maintain a professional, synchronized workflow across diverse digital ecosystems.

Task 4 Draft a Professional Email & Etiquette Guide

Part B – Create a Social Media Do's and Don'ts checklist:

A significant real-world example of poor digital communication is the **2012 Knight Capital Group software glitch**, which cost the company **$440 million** in just 45 minutes. The crisis was triggered when a technician manually deployed new code to only seven of eight servers, leaving the eighth running obsolete software. Because there was no automated deployment log or a clear digital "pre-flight" checklist shared across the team, the discrepancy went unnoticed until the mismatched servers began executing millions of erratic trades.

To prevent this, the team should have utilized **automated deployment pipelines** and a centralized **incident response channel**. Instead of relying on manual oversight, a digital dashboard could have flagged the server version mismatch instantly. Standardizing communication through a shared "Deployment Manifest" would have ensured that every team member had visibility into the system's state, transforming a catastrophic blind spot into a routine technical check.

Task 5 Cybercrime Awareness Case Study & Prevention Guide

Part B – Prevention Checklist:

Stay Safe Online: A Checklist for Students in India

Navigating the digital landscape requires more than just basic awareness; it requires specific, proactive habits. Use this checklist to secure your academic and personal data:

1. **Enable UPI Transaction Limits:** Set a daily limit on your UPI apps (like BHIM, GPay, or PhonePe) to minimize potential loss if your phone is compromised.
2. **Verify Official "Gov.in" Domains:** When applying for scholarships or competitive exams, always ensure the URL ends in .gov.in or .nic.in to avoid sophisticated phishing clones.
3. **Use 2FA on University Portals:** Activate Two-Factor Authentication (2FA) for your college email and student portals to prevent identity theft or unauthorized grade tampering.
4. **Avoid Public Wi-Fi for Banking:** Refrain from accessing banking apps or sensitive student accounts on free railway or campus Wi-Fi without a trusted VPN.
5. **Secure Your Aadhaar Biometrics:** Use the **mAadhaar app** or UIDAI website to "Lock" your biometrics, preventing unauthorized KYC attempts.
6. **Scrutinize WhatsApp Forwarded Links:** Treat any message promising "Free Laptops" or "Job Schemes" from the government with extreme skepticism; never click the included shortened URLs.

7. **Check App Permissions:** Regularly audit permissions for study-aid apps to ensure they aren't accessing your contacts or microphone unnecessarily.
8. **Report Cybercrime Immediately:** Bookmark **cybercrime.gov.in** and know the National Helpline Number (**1930**) to report financial fraud within the "golden hour."

Research Reflection: The Reality of Digital Vulnerability

What surprised me most during this research was the sheer sophistication of **identity-based phishing** targeting Indian students. I previously assumed that most scams were easily detectable through poor grammar; however, the emergence of highly realistic "clone sites" for national examinations and scholarship portals is alarming. These sites don't just steal money; they harvest high-stakes personal data like Aadhaar numbers and academic transcripts, which can be used for long-term identity fraud.

As a result, the one habit I will personally change is my **management of biometrics**. I realized that while I secure my passwords, my biometric data remains "open" for authentication by default. I plan to use the UIDAI portal to lock my Aadhaar biometrics immediately. I will only unlock them temporarily when I am physically present for a legitimate verification, adding a vital layer of security to my most sensitive national identity credential.

Conclusion

The completion of this Digital Literacy project has been a transformative step in my journey as an Integrated M.Tech CSE (Cyber Security) student. Through the systematic execution of each task—from building a professional digital portfolio on GitHub and LinkedIn to analyzing the nuances of secure digital communication—I have moved beyond being a mere consumer of technology to becoming a responsible and proactive digital citizen.

This project highlighted that technical proficiency must be balanced with cyber hygiene and professional ethics. Researching the specific digital threats facing students in India, such as sophisticated phishing and biometric vulnerabilities, has underscored the practical importance of the security principles I am studying in my degree. By implementing tools like Two-Factor Authentication and Aadhaar biometric locking, I have already begun to translate academic knowledge into personal safety.

Ultimately, these activities have provided me with a documented record of my growth and a clear roadmap for the next four years. I am now better equipped to collaborate effectively within a professional ecosystem, communicate technical concepts with clarity, and maintain a secure, professional online presence that reflects my aspirations in the field of Cyber Security.

REFERENCES

GitHub - github.com

LinkedIn- linkedin.com

Kaggle - kaggle.com

HackerRank - hackerrank.com

CodeChef - codechef.com

Google Forms - forms.google.com

Cyber Crime Portal - cybercrime.gov.in

Cyber Crime Helpline 1930 (24x7 phone)